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An EEG Dataset for Visual Imagery-Based Brain–Computer Interface

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With the advancement of non-invasive brain–computer interface (BCI) technologies, decoding high-level cognitive activity has become pivotal for expanding human–machine interaction. Visual imagery-based BCI (VI-BCI) enable voluntary activation of specific brain regions without external cue, offering novel pathways for immersive applications. However, research on the neural representation of such complex cognitive tasks is still limited, and most existing electroencephalogram (EEG) datasets primarily target motor imagery, hindering the development of robust VI decoding models. Here we present an EEG dataset recorded from 22 participants performing visual imagery tasks involving ten commonly recognized images across three categories: figures, animals, and objects. Each participant completed two sessions, with EEG recorded from 32-channels at 1000 Hz. This resource helps overcome data homogeneity issues in VI studies and provides a foundation for exploring neuroplasticity, adaptive decoding algorithms, and cross-subject generalization, facilitating the transition from controlled experiments to real-world applications.

Background & Summary

Brain–computer interface (BCI) technology establishes a direct communication pathway between the human brain and external devices, demonstrating significant potential in fields such as neurorehabilitation, intelligent control, and human–machine interaction^{1,2}. Among various BCI paradigms, visual imagery-based BCI (VI-BCI) have attracted increasing attention in recent years due to their unique advantage of eliciting specific neural activity without external stimulation^{3,4}, enabling voluntary cognitive engagement^{5–8}. By leveraging electroencephalography (EEG), VI-BCI aim to decode neural patterns associated with visual imagery tasks⁹. However, despite their promise, the development of VI-BCI still faces multiple challenges, including limitations in decoding accuracy, inter-subject variability, and the scarcity of high-quality, task-specific datasets^{10–12}.

Currently, numerous open-access EEG datasets are available within the BCI community^{13–17}, most of which are centered around motor imagery (MI) tasks. These MI datasets are relatively mature and have been widely employed for algorithm development and model validation^{18–20}. For example, the BCI Competition IV 2a dataset, which includes EEG data from nine subjects performing four motor imagery tasks (left hand, right hand, feet, and tongue), is frequently used for validation in related studies. In contrast, publicly available datasets targeting VI tasks remain scarce, which hampers the in-depth exploration of VI-BCI systems. This scarcity is largely attributed to the inherently weak and variable neural signals associated with visual imagery, which are highly susceptible to individual differences and fluctuations in cognitive state. Moreover, the low signal-to-noise ratio of EEG signals further complicates the process of reliable feature extraction^{21–23}.

To address these limitations and advance VI-BCI research, we have developed and released a novel EEG dataset specifically designed for visual imagery decoding. This dataset includes EEG recordings from 22 healthy participants performing imagery tasks involving ten commonly encountered images across three semantic categories: figures (circle, square, pentagram), animals (dog, fish, bird), and objects (cup, chair, watch, scissors). The task paradigm was designed with a clear gradient of cognitive complexity, progressing from low-load abstract geometric figures to biologically meaningful animals and finally to complex everyday objects with functional attributes. Geometric figures represent highly abstract and simple visual forms, whereas animals and common objects constitute more concrete and semantically rich imagery. Within the object category, four tasks were included to capture variations in spatial dimensionality: cup and chair are typically perceived as three-dimensional objects, while watch and scissors emphasize two-dimensional planar outlines. Therefore,

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Dataset	Sample size	Task paradigm	Channels	Access method
Wilson <i>et al.</i> ¹⁴	12 participants (3 completed 2 sessions)	3 semantic concepts (penguin, guitar, flower)	128	OpenNeuro repository
Pan <i>et al.</i> ²⁴	6 participants (single session)	29 characters (26 lowercase letters and 3 punctuation marks)	32	Request via Email
This paper	22 participants (19 completed 2 sessions)	Figure (circle, square, pentagram); Animal (dog, fish, bird); Object (cup, chair, watch, scissors)	32	figshare repository

Table 1. Comparison of representative visual imagery EEG datasets. This table summarizes key characteristics of existing visual imagery EEG datasets, including sample size, task paradigm, channels, and access method.

this study not only incorporates a cognitive gradient from abstract to concrete imagery but also systematically examines the influence of spatial attribute differences among objects, aiming to comprehensively characterize the neural representations underlying visual imagery of varying complexity. All selected images depicted common and highly familiar items in daily life, ensuring both the feasibility of imagery tasks and the stability of neural responses.

To further highlight the characteristics of the proposed dataset, we conducted a systematic comparison with several representative visual imagery EEG datasets^{14,24} (see Table 1). The comparison dimensions include sample size, task paradigm, recording configuration, and data accessibility. Existing visual imagery datasets generally involve a small number of participants, limited task categories, and paradigms focused on simple geometric forms or broad semantic concepts. Most datasets contain data from a single session only, making it difficult to evaluate within-subject stability or cross-session consistency. In contrast, our dataset includes EEG recordings from 22 participants performing 10 visual imagery tasks organized along a clear cognitive complexity gradient. The data were recorded using a 32-channel, 1000 Hz EEG system and have been publicly released to facilitate reproducibility and further research. This dataset provides a more representative and scalable foundation for algorithm development and cognitive neuroscience studies.

Overall, this dataset is among the few publicly available EEG resources that focus specifically on VI tasks and provide a relatively large sample size. It fills a critical gap in the VI-BCI field with respect to data accessibility and diversity. We anticipate that this dataset will serve as a valuable foundation for future studies on EEG-based feature modeling, classification algorithm development^{8,25}, and data augmentation strategies, ultimately facilitating the transition of VI-BCI systems from laboratory settings to real-world applications.

Methods

Participants. A total of 22 healthy adult volunteers (aged 20–23 years, including 5 females) participated in this study. All participants were enrolled students at Kunming University of Science and Technology. Among them, two had prior experience with BCI-related experiments or training. All participants had normal or corrected-to-normal vision (corrected visual acuity ≥ 1.0) and no history of neurological disorders, color vision deficiencies, or other conditions that might affect experimental results. To protect personal identity, each participant was coded using a unique subject ID ranging from “sub-01” to “sub-22”. Each participant completed two independent EEG recording sessions using the same VI paradigm, with a minimum interval between sessions to account for intra- and inter-session variability. Two participants did not complete the second session due to personal reasons.

This study was approved by the Medical Ethics Committee of Kunming University of Science and Technology (approval number: KMUST-MEC-2023-029) and conducted in strict accordance with the Declaration of Helsinki. Prior to participation, all individuals received a detailed explanation of the experimental procedures, potential risks, and requirements (such as maintaining sufficient sleep and abstaining from alcohol), as well as instructions on how to perform the VI tasks. After ensuring their comprehension, written informed consent was obtained. Importantly, the informed consent form included an explicit clause stating that participants agreed to allow their EEG data to be anonymized and openly released on the Figshare repository for academic research purposes, and that such data may be accessed and reused by other researchers under an open license. Only fully anonymized data were released, and all identifying information was removed to prevent re-identification. All procedures were carried out with strict privacy protection measures, ensuring voluntary participation and safeguarding participants’ rights.

Questionnaire. The vividness of participants’ visual imagery was assessed using the Vividness of Visual Imagery Questionnaire (VVIQ)²⁶. This instrument consists of 16 items requiring participants to self-rate the clarity of imagined scenarios. Each item is scored on a 5-point Likert scale, with 5 indicating the most vivid imaginable image and 1 indicating no imagery at all. The total score ranges from 16 to 80, with higher scores reflecting stronger visual imagery ability. The average VVIQ score among participants was 61.68, and all individuals scored above the standard threshold for adequate visual imagery ability (>40)²⁷, indicating that the sample population possessed a sufficient baseline of visual cognitive capability.

Experimental paradigm. The experimental paradigm included three categories of VI tasks: figures, animals, and objects. The figures consisted of images of a circle, a square, and a pentagram; the animal category included images of a dog, a fish, and a bird; and the object category contained images of a cup, a chair, a watch, and a pair of scissors. All cue were color images with a resolution of 440×330 pixels, displayed on a white background (see Fig. 1). The paradigm was implemented using Psychtoolbox-3²⁸ and presented on a screen with a resolution of 1920×1080 pixels.



Fig. 1 Visual imagery cue used in the experiment, categorized into geometric figures, animals, and everyday objects.

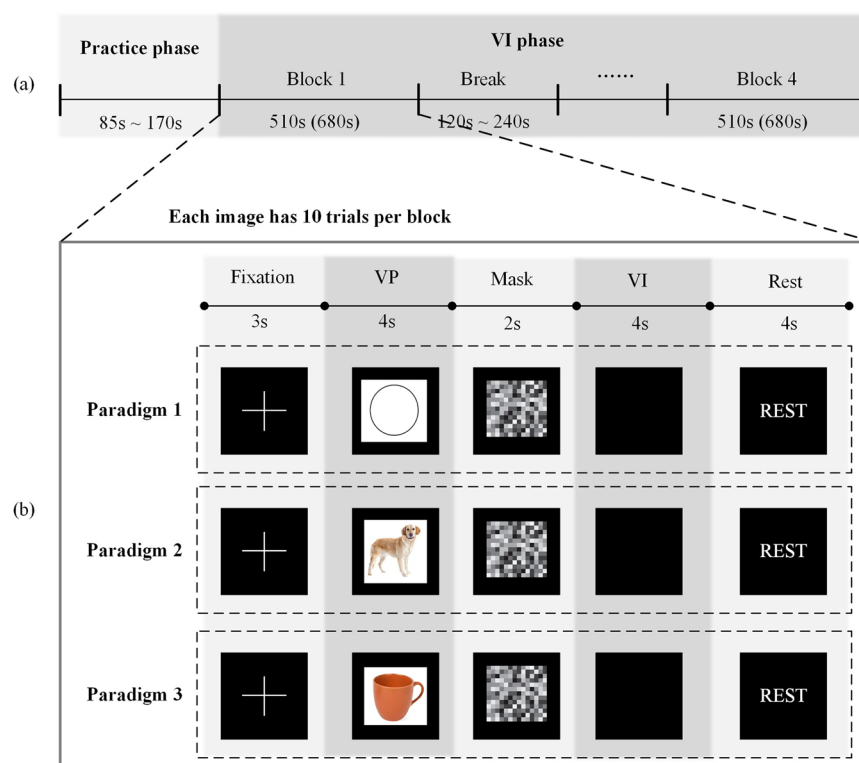


Fig. 2 Timing diagram of VI-BCI paradigm. **(a)** The experimental paradigm included a rest period and a VI period, with the VI period consisting of four VI blocks. **(b)** The experimental paradigm in a single trial.

To ensure the consistency of visual cue, the brightness, contrast, and color saturation of all images were standardized. All cue were normalized to have comparable mean luminance and contrast levels, minimizing low-level visual feature differences across categories and thereby enhancing the reproducibility of the experimental paradigm.

Each recording session lasted between 37 and 56 minutes and comprised four VI blocks for each task category. Flexible rest periods of 2 to 4 minutes were provided between blocks to mitigate the effects of fatigue. In the figure and animal categories, each image was presented 40 times, yielding a total of 120 trials per category. Each trial lasted 17 seconds and followed a structured sequence (see Fig. 2). The trial began with the presentation of a central fixation cross for 3 seconds to prompt participants to remain alert and relaxed. This was followed by the random presentation of a task image for 4 seconds, during which participants were instructed to observe and memorize the image. After image offset, a visual mask was displayed briefly to eliminate residual visual aftereffects¹⁴. Then, the screen turned black, signaling the start of the VI task, during which participants performed mental imagery for 4 seconds. A “Rest” cue then appeared for 4 seconds, during which participants were instructed to stop imagery and rest.

Throughout the experiment, no unnecessary auditory or visual cue were presented. As illustrated in Fig. 2, each VI block consisted of either 30 or 40 trials. Specifically, the figure imagery task included 120 trials in total, the animal imagery task included 120 trials, and the object imagery task consisted of 160 trials.

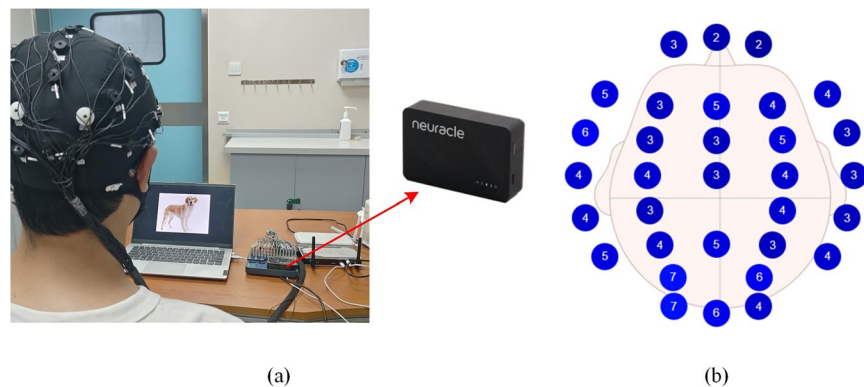


Fig. 3 The data acquisition equipment used in the experiment. **(a)** EEG cap and acquisition system. **(b)** Electrode positions and impedance values on an EEG cap worn by a participant.

Data acquisition. Each participant wore a 32-channel EEG cap, and EEG signals were recorded using a wireless EEG equipment independently developed by Neuracle, as shown in Fig. 3(a). Task labels were synchronized via serial port communication to ensure precise alignment between task events and EEG signals. Electrodes were arranged according to the extended international 10–20 system, with 32 electrodes actively recording EEG at a sampling rate of 1000 Hz. The AFz electrode served as ground, and Cpz was used as the reference. This configuration followed the specifications of the wireless EEG device manufacturer to optimize signal quality. Moreover, the mid-line position of Cpz provided a stable reference for imagery-related parietal activity while minimizing lateral bias.

All experiments were conducted in a soundproof and light-isolated room to ensure stable ambient lighting and minimize external noise interference. Before the main experiment, participants completed VVIQ to assess their imagery ability. EEG preparation included placing the cap and applying conductive paste to reduce electrode impedance below 5kΩ, which took approximately 30 to 50 minutes. Participants then completed 5 to 10 practice trials to familiarize themselves with the experimental procedure and task demands.

Impedance values for each electrode channel were visualized prior to recording (see Fig. 3(b)); the numeric labels within each circle indicate channel impedance in kΩ. Once preparations were complete, participants could initiate the task by pressing the space bar on the keyboard. Upon completion of the experiment, EEG data recording was stopped, and all EEG signals along with event markers were saved in .bdf format for further analysis.

Data Pre-processing. The EEG signals were preprocessed using the MNE-Python library²⁹ to improve signal quality and minimize artifacts. Initially, the raw EEG data in .bdf format were loaded and the electrodes were configured according to the international 10–10 standard system. A common average reference was then applied.

To suppress noise, a band-pass filter between 5–30 Hz was implemented using a Hamming-windowed FIR filter with zero-phase filtering and edge padding to reduce boundary effects.

Artifact correction was performed via Independent Component Analysis (ICA) using the Picard³⁰ algorithm, which provides faster convergence and improved numerical stability compared to conventional ICA methods such as FastICA, making it particularly suitable for high-density EEG data. Before ICA decomposition, EEG data were band-pass filtered between 5–30 Hz using a zero-phase Hamming-window FIR filter and re-referenced to the common average. The ICA decomposition was performed with 25 components and a maximum of 300 iterations to ensure stable convergence. ICA decomposition results were automatically labeled using the ICLabel³¹ method, which categorizes components into “brain,” “muscle,” “eye,” “heart,” “line noise,” “channel noise,” and “other.” Only components classified as “brain” or “other” were retained for the reconstruction of EEG signals.

For epoching, 4-second segments were extracted from the EEG data starting at event markers corresponding to different visual imagery tasks. Epochs were extracted from 0–4 seconds post-stimulus, with a pre-stimulus baseline correction window of –0.2 – 0 s. Each epoch was baseline-corrected and stored in both .mat and .tsv formats for subsequent analysis. A final quality check was performed to ensure that each subject retained a balanced number of trials per class, and trial information was documented in accompanying metadata files.

Data Records

The dataset is available at Figshare³² repository, with this section being the primary source of information on the availability and content of the data being described. The file structure and naming follows Brain Imaging Data Structure (BIDS) format³³. Figure 4 illustrates the directory structure for the EEG data in BIDS format, where only one participant is shown for clarity.

Each subject folder (e.g., sub-01) contains one or two sessions (ses-01, ses-02), depending on the availability of data. Within each session, the eeg/ subdirectory stores raw EEG signals in the BIDS-compatible (*.bdf) formats, accompanied by metadata files such as channel information (*.channels.tsv), event markers (*.events.tsv), and JSON sidecar files (*.eeg.json). This organization ensures consistency across participants and allows straightforward integration with common EEG analysis toolboxes.

It should be noted that two participants (sub-09 and sub-10) only completed the first session (ses-01) due to personal circumstances, and therefore have no available data for the second session (ses-02). In addition, the animal

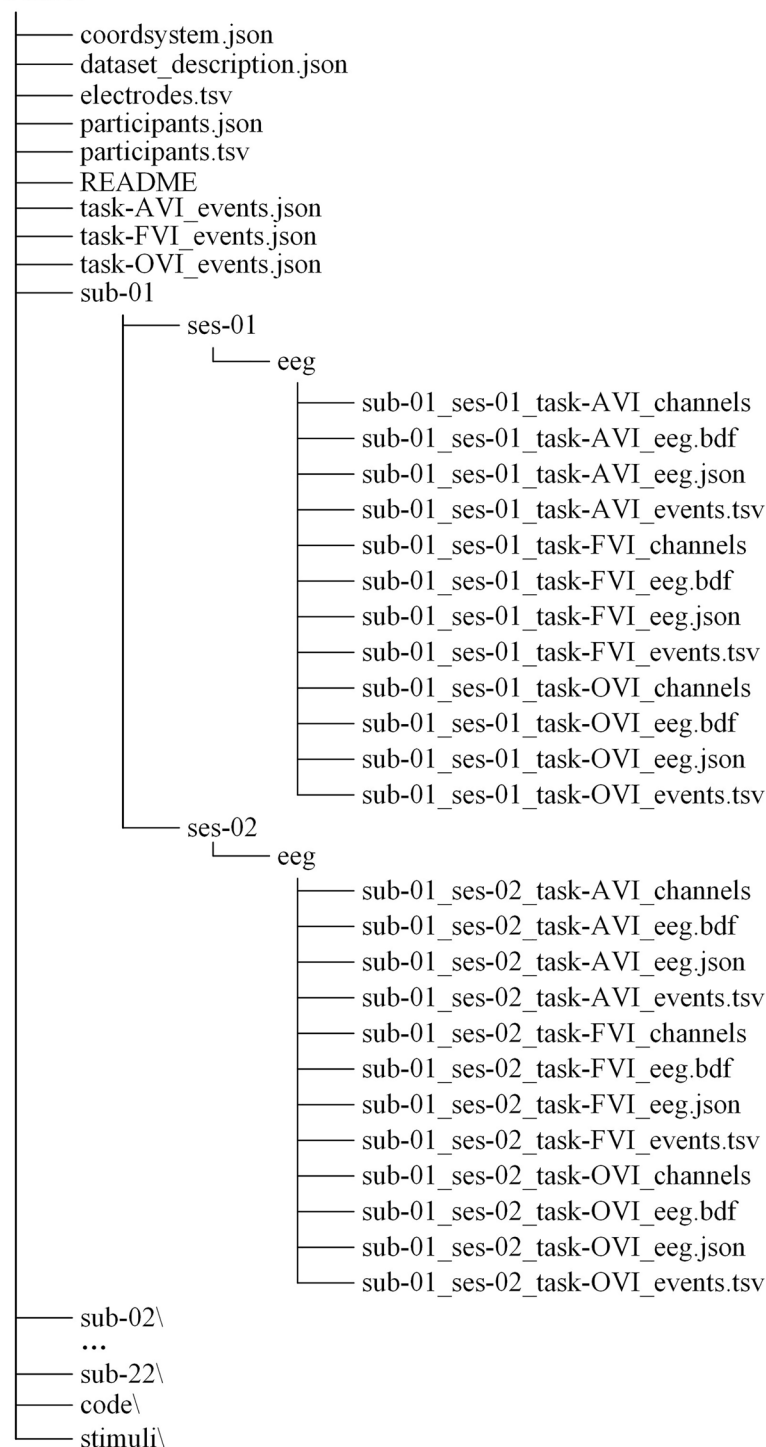
VI-EEG

Fig. 4 The directory structure of the data according to BIDS format. Two versions of the EEG data are provided, raw and pre-processed versions.

imagery data of sub-08 in ses-02 was excluded from analysis because of poor signal quality, which introduced substantial noise contamination and led to unreliable classification performance.

Technical Validation

To evaluate the decodability of the proposed visual imagery EEG dataset, we conducted classification experiments using two representative approaches: (1) the traditional spatial filtering method Common Spatial Pattern (CSP)¹⁸ combined with k-nearest neighbor (KNN) classification, serving as a classical baseline; and (2) the convolutional neural network EEGNet³⁴, specifically designed for EEG-based brain-computer interface

Model	Data Splitting Strategy	Optimizer	Learning Rate	Batch Size	Epochs	Classifier Settings
EEGNet	Within-subject stratified split (80% train, 20% test)	Adam	0.001	64	500	Cross-entropy loss
CSP + KNN	Within-subject split, test_size = 0.2, random_state = 42	—	—	—	—	CSP reg = "OAS"; KNN n_neighbors = 5; weights = "distance"

Table 2. Data splitting and model configuration.

		Animal		Figure		Object	
		session1	session2	session1	session2	session1	session2
CSP	F1-score	0.5014	0.4871	0.5047	0.4864	0.411525	0.4164
	Recall	0.5017	0.4877	0.5054	0.4875	0.411875	0.4175
	Precision	0.5096	0.4974	0.5125	0.4940	0.41685	0.4228
	Accuracy	0.5017	0.4877	0.5054	0.4875	0.411875	0.4175
EEGNet	F1-score	0.7661	0.7440	0.7580	0.7433	0.6155	0.6183
	Recall	0.7658	0.7439	0.7571	0.7433	0.6172	0.6181
	Precision	0.7764	0.7527	0.7687	0.7531	0.6258	0.6259
	Accuracy	0.7658	0.7439	0.7571	0.7433	0.6172	0.6182

Table 3. Classification results of different methods on three visual imagery tasks. The table presents the average classification performance of CSP and EEGNet methods on animal, figure, and object visual imagery tasks, evaluated separately across two sessions.

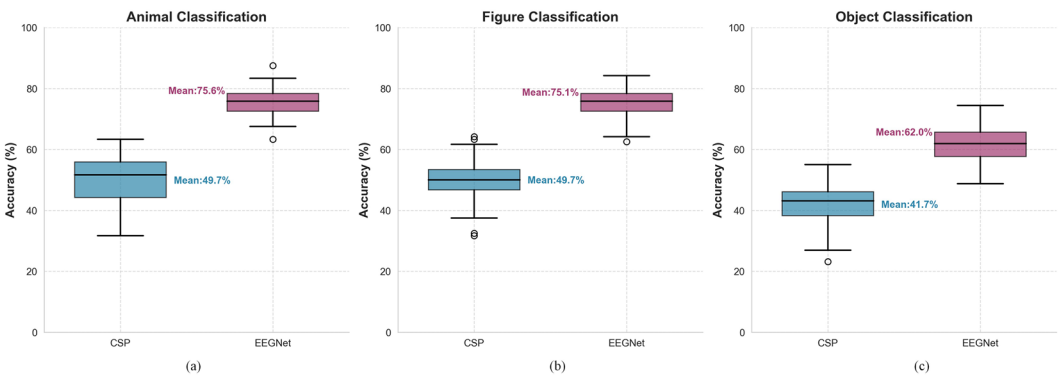


Fig. 5 Comparison of classification accuracy across three methods for different visual imagery tasks. (a) Classification accuracy of the animal imagery task, (b) classification accuracy of the figure imagery task, and (c) classification accuracy of the object imagery task.

applications. CSP extracts discriminative features by maximizing variance differences between conditions and, when paired with a simple classifier, provides a robust benchmark. In contrast, EEGNet integrates temporal, spatial, and depthwise separable convolutions, enabling efficient modeling of both local and global EEG patterns while maintaining a compact architecture.

To ensure fair and reproducible model evaluation, a within-subject, session-wise validation strategy was employed for all classification analyses. For each participant, the data were split into 80% training and 20% testing sets using stratified sampling to maintain consistent class proportions across subsets. This design reflects intra-subject decoding performance while minimizing the influence of inter-subject variability. We list the key hyperparameter configurations of the data partitioning strategy and classification model (see Table 2).

It should be clarified that all classification accuracies reported in this study refer to the discriminative performance within the same semantic category (e.g., among dog, fish, and bird within the animal category). This evaluation is intended to verify the dataset’s feasibility for fine-grained visual imagery decoding, rather than for assessing inter-category distinctions across broader semantic classes.

As shown in Table 3, EEGNet consistently outperformed CSP across all three visual imagery tasks (animals, figures, and objects). Specifically, EEGNet achieved average accuracies of 75.8% for animal imagery and 75.1% for figure imagery, representing improvements of approximately 26 percentage points over CSP. For object imagery, EEGNet reached an average accuracy of 62.0%, exceeding CSP by more than 20 percentage points. Beyond higher accuracy, EEGNet exhibited markedly reduced variability across participants, with narrower interquartile ranges and fewer outliers (see Fig. 5), indicating improved generalization and stability.

The results across two experimental sessions further demonstrate the reliability of the dataset and paradigm. As illustrated in Fig. 6, the average accuracy differences between Session 1 and Session 2 were all within 2.5%,

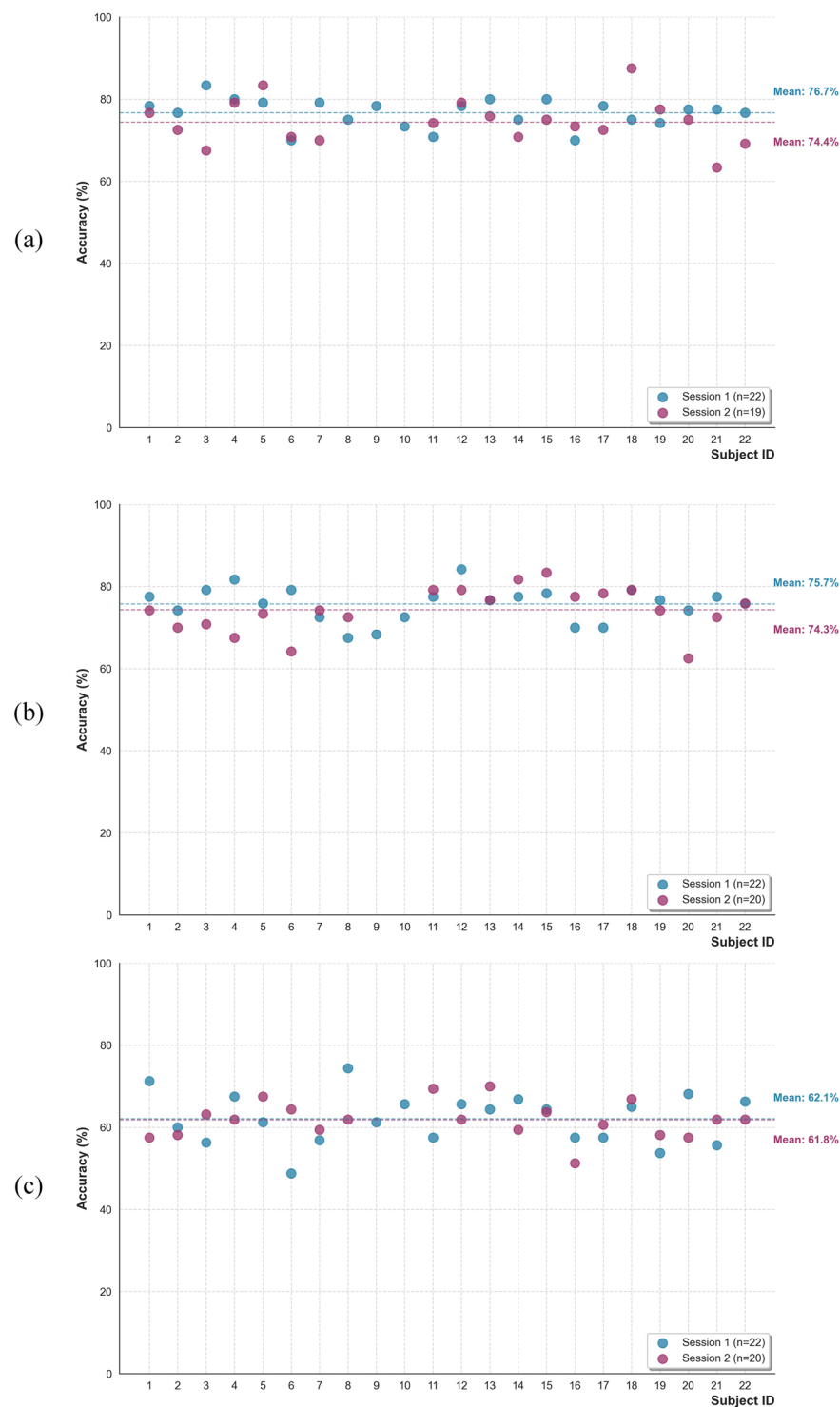


Fig. 6 Comparison of classification accuracies between session 1 and session 2 for three visual imagery tasks: (a) animal imagery, (b) figure imagery, and (c) object imagery. Results show consistent accuracy distributions across sessions and a clear task difficulty gradient.

and most participants showed comparable performance across sessions, suggesting stable decoding of visual imagery EEG signals over time.

Confusion matrix analysis (see Fig. 7) further confirmed robust classification performance, with clear diagonal dominance observed across all three tasks. Importantly, the misclassification patterns were structured and cognitively interpretable rather than random. For example, frequent confusions occurred between dog and bird in the animal task, reflecting perceptual proximity; in the figure task, occasional confusions occurred between square and pentagram due to similar geometric features; and in the object task, between watch and scissors due

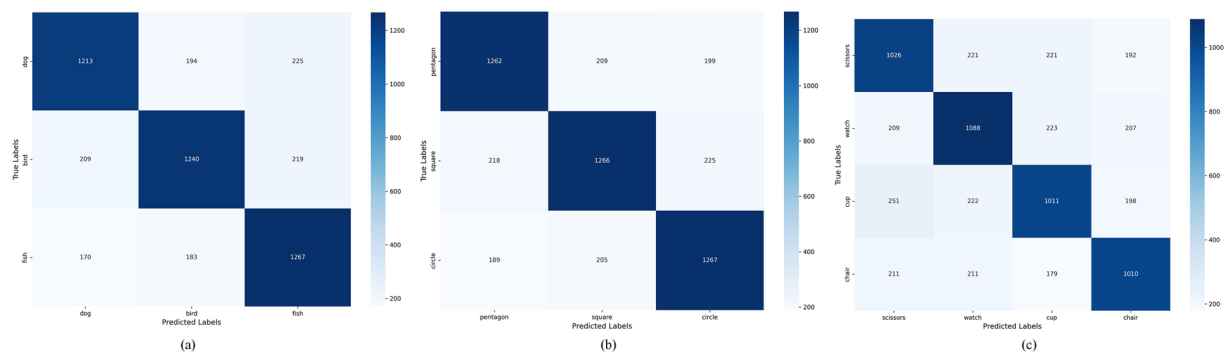


Fig. 7 Confusion matrices of EEGNet classification results for three visual imagery tasks: **(a)** animal imagery, **(b)** figure imagery, and **(c)** object imagery. Strong diagonal dominance demonstrates excellent classification ability, while structured misclassification patterns reflect meaningful cognitive similarities across categories.

to fine structural similarities. These systematic error patterns align with human cognitive characteristics, further validating both the cognitive plausibility and technical reliability of the dataset.

In summary, the dataset shows good decodability and stability when validated with two representative methods. EEGNet clearly outperforms the traditional CSP approach, and the interpretable misclassification patterns further confirm the dataset's reliability and research value.

Usage Notes

Users are encouraged to preprocess the data according to their own analysis pipelines. Standard EEG processing methods such as ICA artifact removal, bandpass filtering, and epoching based on event markers are recommended. The dataset is suitable for studies on EEG feature extraction, classification algorithm development, and visual imagery decoding. When publishing results, please cite this dataset accordingly.

Data availability

The dataset supporting this study has been deposited in Figshare and is publicly available at <https://doi.org/10.6084/m9.figshare.30227503>. No additional restrictions apply to the data.

Code availability

Sample scripts for data loading, preprocessing, and a basic classification example are included in the Figshare repository of this dataset, under the code/ directory. The repository also provides full environment specifications to ensure reproducibility, with the main dependencies being: Python 3.9.21, MNE 1.8.0, mne-icalabel 0.7.0, and PyTorch 2.5.0. This code provides a baseline implementation for further analyses and can be readily adapted for various machine learning and deep learning approaches. The repository is available at <https://doi.org/10.6084/m9.figshare.30227503>.

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Competing interests

The authors declare no competing interests.

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